

Lecture 7: The first fundamental form

Recall.

Definition. A subset $S \subset \mathbb{R}^3$ is a smooth **surface**, or an embedded two-dimensional submanifold, if every point $p \in S$ has a neighborhood in which S is the image of an embedding

$$f : U \subset \mathbb{R}^2 \longrightarrow \mathbb{R}^3.$$

Equivalently, for each $p \in S$ there is an open neighborhood $W \subset \mathbb{R}^3$ and a smooth function $F : W \rightarrow \mathbb{R}$ such that

$$S \cap W = F^{-1}(0), \quad \nabla F \neq 0 \text{ on } F^{-1}(0).$$

Thus:

- a parametrized surface element is a local description;
- an immersion may have global self-intersections;
- a submanifold is locally embedded and is covered by compatible parametrizations.

Both definitions describe a surface that is locally a graph. Fix $p \in S$; shrink neighborhoods as needed.

Starting with an embedding f , the rank of Df is two. After relabeling coordinates, $G = (f_1, f_2)$ has invertible derivative at $f^{-1}(p)$. The inverse function theorem gives a smooth local inverse for G , so

$$f \circ G^{-1}(x, y) = (x, y, h(x, y)).$$

Because f is an embedding, we can choose an ambient neighborhood W in which S is exactly this graph. Set

$$F(x, y, z) = z - h(x, y), \quad \nabla F = (-h_x, -h_y, 1) \neq 0.$$

Then $S \cap W = F^{-1}(0)$.

Conversely, suppose $S \cap W = F^{-1}(0)$ and $\nabla F(p) \neq 0$. Relabel coordinates so that $F_z(p) \neq 0$, and set

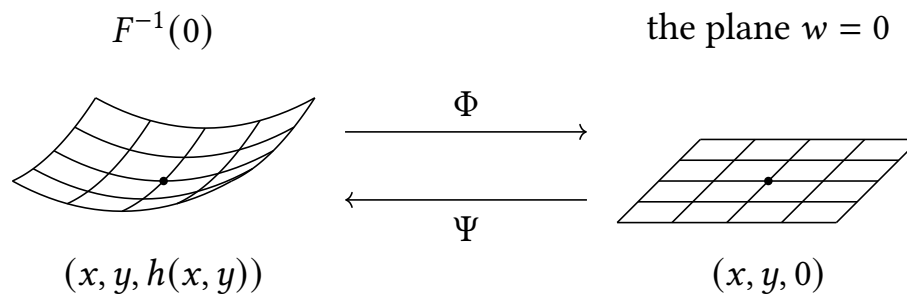
$$\Phi(x, y, z) = (x, y, F(x, y, z)).$$

Since $\det D\Phi(p) = F_z(p) \neq 0$, the inverse function theorem gives a smooth local inverse Ψ . The zero set $F^{-1}(0)$ is parametrized by

$$(x, y) \longmapsto \Psi(x, y, 0) = (x, y, h(x, y)).$$

This is an embedding: its derivative has rank two, and projection onto (x, y) is its continuous inverse.

The change of coordinates Φ flattens the surface locally. Its inverse Ψ recovers the graph.



The new third coordinate is $w = F(x, y, z)$. Thus

$$\Phi(x, y, z) = (x, y, F(x, y, z)), \quad \det D\Phi(p) = F_z(p) \neq 0.$$

The inverse function theorem gives $\Psi = \Phi^{-1}$ locally. On the plane $w = 0$,

$$\Psi(x, y, 0) = (x, y, h(x, y)), \quad F(x, y, h(x, y)) = 0.$$

Here $h(x, y)$ is the height of the surface: the third component of $\Psi(x, y, 0)$. The first two coordinates stay unchanged.

In the other direction, a graph is a regular level set of $F(x, y, z) = z - h(x, y)$, since $F_z = 1$.

Lemma. Let $F : W \subset \mathbb{R}^3 \rightarrow \mathbb{R}$ be smooth, with W open, and let $S = F^{-1}(0)$. If $p \in S$ and $\nabla F(p) \neq 0$, then S is a surface near p , and

$$v \in T_p S \iff dF_p(v) = 0.$$

In other words, $T_p S = \ker dF_p$.

Proof. If $v \in T_p S$, choose a smooth curve $c(t)$ in S with $c(0) = p$ and $\dot{c}(0) = v$. Since $F(c(t)) = 0$, the chain rule gives

$$0 = \left. \frac{d}{dt} F(c(t)) \right|_{t=0} = dF_p(\dot{c}(0)) = dF_p(v).$$

Thus $T_p S \subseteq \ker dF_p$.

Since $\nabla F(p) \neq 0$, the linear map $dF_p : \mathbb{R}^3 \rightarrow \mathbb{R}$ has rank one. By rank-nullity,

$$\dim \ker dF_p = 3 - 1 = 2 = \dim T_p S.$$

The inclusion of these two-dimensional spaces is therefore equality.

Finally, $dF_p(v) = \langle \nabla F(p), v \rangle$, so

$$T_p S = \nabla F(p)^\perp.$$

Thus the gradient is a normal vector to the surface.

For example, the sphere

$$S_R^2 = \{(x, y, z) : x^2 + y^2 + z^2 = R^2\}$$

is a submanifold. Set

$$F(x, y, z) = x^2 + y^2 + z^2 - R^2.$$

Then

$$\nabla F(x, y, z) = 2(x, y, z),$$

which never vanishes on S_R^2 .

At $p \in S_R^2$, a tangent vector X satisfies

$$dF_p(X) = 2\langle p, X \rangle = 0.$$

Therefore

$$T_p S_R^2 = p^\perp,$$

and the radial vector p is normal to the sphere.

This description agrees with the tangent space obtained from any local parametrization.

Examples. The cylinder of radius $R > 0$,

$$S = \{(x, y, z) : x^2 + y^2 = R^2\},$$

is an embedded surface. For $H(x, y, z) = x^2 + y^2 - R^2$,

$$\nabla H = (2x, 2y, 0) \neq 0 \quad \text{on } S.$$

At $p = (x, y, z) \in S$,

$$T_p S = \{(a, b, c) : xa + yb = 0\}.$$

The radial vector $(x, y, 0)$ is normal; the vertical direction $(0, 0, 1)$ is tangent everywhere.

The saddle $S = \{z = xy\}$ is also embedded. Here

$$H(x, y, z) = z - xy, \quad \nabla H = (-y, -x, 1) \neq 0.$$

At $p = (u, v, uv)$, its tangent space is

$$T_p S = \{(a, b, c) : c = va + ub\}.$$

Its graph parametrization $f(u, v) = (u, v, uv)$ is a global embedding: projection to the first two coordinates is its inverse. We will compute both surfaces' first fundamental forms below.

The Euclidean inner product in $\mathbb{R}^3$ measures lengths and angles of tangent vectors. To an embedded surface S we obtain a natural inner product on $T_p S$ for every $p \in S$.

Definition. The *first fundamental form* of a parametrized surface element $f : U \rightarrow \mathbb{R}^3$ is

$$I_u(\xi, \eta) = \langle Df_u(\xi), Df_u(\eta) \rangle, \quad \xi, \eta \in \mathbb{R}^2.$$

It is the pullback of the Euclidean inner product by f .

Equivalently, on the tangent space itself,

$$I_p(X, Y) = \langle X, Y \rangle_{\mathbb{R}^3}, \quad X, Y \in T_p S.$$

Because Df_u is injective, I_u is positive definite. It is therefore an inner product on each tangent space.

Let $A : V \rightarrow W$ be linear, and let g be an inner product on W . Its ***pullback*** to V is the bilinear form

$$(A^*g)(v, w) = g(Av, Aw), \quad v, w \in V.$$

We measure vectors in V by first mapping them into W . This form is symmetric, and

$$(A^*g)(v, v) = g(Av, Av) \geq 0,$$

with equality exactly when $Av = 0$. Thus A^*g is an inner product if and only if A is injective.

Now let $\varphi : M \rightarrow N$ be a smooth map, with M, N surfaces or open subsets of Euclidean spaces. A ***metric*** g on N is a smoothly varying inner product g_p on each T_pN . At each $p \in M$, apply the same construction to the linear map $d\varphi_p : T_pM \rightarrow T_{\varphi(p)}N$:

$$(\varphi^*g)_p(v, w) = g_{\varphi(p)}(d\varphi_p(v), d\varphi_p(w)).$$

This is a metric on M when φ is an immersion. Otherwise, a nonzero vector in $\ker d\varphi_p$ has zero squared length for the pullback.

For a parametrized surface $f : U \rightarrow \mathbb{R}^3$, pulling back the Euclidean metric gives precisely the first fundamental form:

$$I = f^*\langle \cdot, \cdot \rangle_{\mathbb{R}^3}.$$

Relative to the coordinate basis (f_u, f_v) , the metric is represented by the matrix of their pairwise inner products. Writing

$$E = \langle f_u, f_u \rangle, \quad F = \langle f_u, f_v \rangle, \quad G = \langle f_v, f_v \rangle,$$

and using $\langle f_v, f_u \rangle = \langle f_u, f_v \rangle$, we obtain the symmetric matrix of the first fundamental form:

$$g = \begin{pmatrix} E & F \\ F & G \end{pmatrix} = (Df)^\top Df.$$

For $\xi = (a, b)$,

$$I_u(\xi, \xi) = Ea^2 + 2Fab + Gb^2.$$

One often writes the same quadratic form symbolically as

$$ds^2 = E du^2 + 2F du dv + G dv^2.$$

The symbols du, dv record components of tangent vectors; they are not small finite displacements.

The immersion condition appears directly in the first fundamental form. By the cross-product identity,

$$\|f_u \times f_v\|^2 = \|f_u\|^2 \|f_v\|^2 - \langle f_u, f_v \rangle^2.$$

Thus

$$\det g = EG - F^2 = \|f_u \times f_v\|^2 > 0.$$

Consequently,

$$E > 0, \quad G > 0, \quad EG - F^2 > 0.$$

The area of the parallelogram spanned by f_u and f_v is

$$\sqrt{EG - F^2}.$$

This leads to the surface-area element

$$dA = \sqrt{EG - F^2} \, du \, dv.$$

Let $\gamma(t) = (u(t), v(t))$ be a curve in the parameter domain and let

$$c(t) = f(\gamma(t))$$

be the corresponding curve on the surface. Then

$$\dot{c} = f_u \dot{u} + f_v \dot{v}.$$

Therefore

$$\|\dot{c}(t)\|^2 = E\dot{u}^2 + 2F\dot{u}\dot{v} + G\dot{v}^2 = I_{\gamma(t)}(\dot{\gamma}, \dot{\gamma}).$$

The length of c is consequently

$$L(c) = \int_a^b \sqrt{E\dot{u}^2 + 2F\dot{u}\dot{v} + G\dot{v}^2} dt.$$

Thus the first fundamental form contains exactly the information needed to measure lengths of curves on the surface.

The first fundamental form also measures angles. For nonzero tangent vectors X, Y ,

$$\cos \theta = \frac{I(X, Y)}{\sqrt{I(X, X)}\sqrt{I(Y, Y)}}.$$

In particular, the coordinate curves meet orthogonally precisely when

$$F = \langle f_u, f_v \rangle = 0.$$

Remarks.

- The first fundamental form depends only on first derivatives of the parametrization.
- It records lengths, angles, and areas within the surface.
- It is not generally the standard inner product on the parameter domain U .
- Its coefficients E, F, G depend on the chosen coordinates, although the geometry they describe does not.

For the graph

$$f(u, v) = (u, v, h(u, v)),$$

we found

$$f_u = (1, 0, h_u), \quad f_v = (0, 1, h_v).$$

Hence

$$E = 1 + h_u^2, \quad F = h_u h_v, \quad G = 1 + h_v^2.$$

The first fundamental form is

$$ds^2 = (1 + h_u^2) du^2 + 2h_u h_v du dv + (1 + h_v^2) dv^2.$$

Even though the parameter domain is flat, the graph generally stretches lengths in a way that depends on the derivatives of h .

Example: the saddle. For the embedded graph $f(u, v) = (u, v, uv)$,

$$f_u = (1, 0, v), \quad f_v = (0, 1, u).$$

Taking the three inner products gives

$$E = 1 + v^2, \quad F = uv, \quad G = 1 + u^2.$$

Thus

$$ds^2 = (1 + v^2) du^2 + 2uv du dv + (1 + u^2) dv^2,$$

and

$$EG - F^2 = 1 + u^2 + v^2 > 0.$$

For instance, at $f(1, 1) = (1, 1, 1)$, the coordinate tangent vectors are $(1, 0, 1)$ and $(0, 1, 1)$. Each has length $\sqrt{2}$, and

$$\cos \alpha = \frac{F}{\sqrt{EG}} = \frac{1}{2}.$$

They meet at angle $\pi/3$, even though the parameter axes meet at a right angle.

The tangent vector with coordinate components $(1, -1)$ is $f_u - f_v = (1, -1, 0)$. Its squared length is

$$E - 2F + G = 2,$$

agreeing with its Euclidean squared length in $\mathbb{R}^3$.

Consider the unit sphere parametrized by longitude and latitude:

$$f(\varphi, \vartheta) = (\cos \varphi \cos \vartheta, \sin \varphi \cos \vartheta, \sin \vartheta).$$

Take $-\pi < \varphi < \pi$ and $-\pi/2 < \vartheta < \pi/2$ for an embedded coordinate patch. Differentiating gives

$$\begin{aligned} f_\varphi &= (-\sin \varphi \cos \vartheta, \cos \varphi \cos \vartheta, 0), \\ f_\vartheta &= (-\cos \varphi \sin \vartheta, -\sin \varphi \sin \vartheta, \cos \vartheta). \end{aligned}$$

Their inner products are

$$E = \cos^2 \vartheta, \quad F = 0, \quad G = 1.$$

Thus

$$ds^2 = \cos^2 \vartheta d\varphi^2 + d\vartheta^2.$$

A change $d\varphi$ in longitude has physical length $\cos \vartheta |d\varphi|$. Circles of latitude shrink toward the poles.

At $\vartheta = \pm\pi/2$, the coefficient E vanishes and the parametrization is not an immersion. The sphere is smooth there; this particular coordinate system is singular.

Example: the cylinder. For the cylinder $x^2 + y^2 = R^2$, use the embedded patch

$$f(\theta, z) = (R \cos \theta, R \sin \theta, z), \quad -\pi < \theta < \pi, \quad z \in \mathbb{R}.$$

This patch omits one vertical line. Other angular intervals give patches covering that line as well. The coordinate tangent vectors are

$$f_\theta = (-R \sin \theta, R \cos \theta, 0), \quad f_z = (0, 0, 1).$$

Consequently,

$$E = R^2, \quad F = 0, \quad G = 1, \quad ds^2 = R^2 d\theta^2 + dz^2.$$

Putting $\ell = R\theta$ gives $ds^2 = d\ell^2 + dz^2$, the Euclidean metric on a strip. Thus unrolling a cylinder preserves lengths and angles locally, although the cylinder bends in $\mathbb{R}^3$.

For a concrete length calculation, consider the helix

$$c(t) = (R \cos t, R \sin t, bt).$$

Its parameter velocity is $(\dot{\theta}, \dot{z}) = (1, b)$, so

$$\|\dot{c}(t)\|^2 = R^2 + b^2, \quad L(c|_{[0, 2\pi]}) = 2\pi\sqrt{R^2 + b^2}.$$

One full turn passes through more than one coordinate patch; the same local formula computes its length throughout.

Example: a cone without its vertex. Fix $a > 0$. The upper cone

$$S = \{(x, y, z) : z = a\sqrt{x^2 + y^2}, z > 0\}$$

is an embedded surface: it is a smooth graph over $\mathbb{R}^2 \setminus \{0\}$. A polar coordinate patch is

$$f(r, \theta) = (r \cos \theta, r \sin \theta, ar), \quad r > 0, \quad -\pi < \theta < \pi.$$

Since

$$f_r = (\cos \theta, \sin \theta, a), \quad f_\theta = (-r \sin \theta, r \cos \theta, 0),$$

we get

$$E = 1 + a^2, \quad F = 0, \quad G = r^2,$$

and therefore

$$ds^2 = (1 + a^2) dr^2 + r^2 d\theta^2, \quad dA = r\sqrt{1 + a^2} dr d\theta.$$

A radial segment from r_0 to $r_1 > r_0$ has length $\sqrt{1 + a^2}(r_1 - r_0)$; a horizontal circle has circumference $2\pi r$.

Adding the vertex does not give a smooth surface there. The ray directions $(\cos \theta, \sin \theta, a)$ cannot all lie in one two-dimensional tangent plane at the vertex.

We now verify that the first fundamental form is independent of the chosen parametrization.

Lemma. Let

$$\tilde{f} = f \circ \phi, \quad \phi : \tilde{U} \longrightarrow U$$

be a change of coordinates. If g and $\tilde{g}$ are the matrices of the first fundamental forms of f and $\tilde{f}$, then

$$\boxed{\tilde{g}(q) = D\phi(q)^{\top} g(\phi(q)) D\phi(q).}$$

Equivalently, for $\xi, \eta \in \mathbb{R}^2$,

$$\tilde{I}_q(\xi, \eta) = I_{\phi(q)}(D\phi_q \xi, D\phi_q \eta).$$

The coefficients change, but the inner product on geometric tangent vectors does not.

Proof. The chain rule gives

$$D\tilde{f}_q = Df_{\phi(q)}D\phi_q.$$

Therefore

$$\begin{aligned}\tilde{\mathbf{I}}_q(\xi, \eta) &= \langle D\tilde{f}_q\xi, D\tilde{f}_q\eta \rangle \\ &= \langle Df_{\phi(q)}D\phi_q\xi, Df_{\phi(q)}D\phi_q\eta \rangle \\ &= \mathbf{I}_{\phi(q)}(D\phi_q\xi, D\phi_q\eta).\end{aligned}$$

In matrix notation,

$$\begin{aligned}\tilde{g} &= (D\tilde{f})^\top D\tilde{f} \\ &= (D\phi)^\top (Df)^\top Df (D\phi) \\ &= (D\phi)^\top g(D\phi).\end{aligned}$$

This proves the lemma.

Example: the paraboloid in two coordinate systems. Return to the paraboloid from the earlier example,

$$f(u, v) = (u, v, u^2 + v^2).$$

The graph formula gives

$$g(u, v) = \begin{pmatrix} 1 + 4u^2 & 4uv \\ 4uv & 1 + 4v^2 \end{pmatrix}.$$

On the portion $u > 0$, introduce polar coordinates:

$$\tilde{f}(r, \theta) = (r \cos \theta, r \sin \theta, r^2), \quad r > 0, \quad -\pi/2 < \theta < \pi/2.$$

Direct differentiation gives

$$\tilde{f}_r = (\cos \theta, \sin \theta, 2r), \quad \tilde{f}_\theta = (-r \sin \theta, r \cos \theta, 0).$$

Thus

$$\tilde{E} = 1 + 4r^2, \quad \tilde{F} = 0, \quad \tilde{G} = r^2,$$

and

$$ds^2 = (1 + 4r^2) dr^2 + r^2 d\theta^2.$$

The polar coordinate curves are orthogonal. In Cartesian coordinates, the cross term $F = 4uv$ is usually nonzero. These are different coordinate grids on the same surface.

Verifying the coordinate-change formula.

For $\phi(r, \theta) = (r \cos \theta, r \sin \theta)$, set

$$A = D\phi = \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix}, \quad w = \begin{pmatrix} r \cos \theta \\ r \sin \theta \end{pmatrix}.$$

The Cartesian metric, evaluated at $\phi(r, \theta)$, is

$$g \circ \phi = I_2 + 4ww^\top.$$

The columns of A are orthogonal, and direct multiplication gives

$$A^\top A = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix}, \quad A^\top w = \begin{pmatrix} r \\ 0 \end{pmatrix}.$$

Hence the lemma predicts

$$\begin{aligned} \widetilde{g} &= A^\top (I_2 + 4ww^\top) A \\ &= \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix} + 4 \begin{pmatrix} r^2 & 0 \\ 0 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 1 + 4r^2 & 0 \\ 0 & r^2 \end{pmatrix}, \end{aligned}$$

exactly as in the direct computation.

In particular, a vector with polar components (a, b) has squared length $(1 + 4r^2)a^2 + r^2b^2$. Its Cartesian components are $A(a, b)^\top$; using g gives the same squared length.

We justify the “area element” introduced earlier.

Lemma. Surface integrals with respect to the area element are independent of the parametrization.

Let $f : U \rightarrow S \subset \mathbb{R}^3$ be an embedding onto a surface patch, and let $\phi : \tilde{U} \rightarrow U$ be a diffeomorphism. Write $\tilde{f} = f \circ \phi$, with metric matrices g and $\tilde{g}$, respectively.

For a continuous function $a : S \rightarrow \mathbb{R}$ with compact support, define

$$\int_S a \, dA = \iint_U a(f(u, v)) \sqrt{\det g(u, v)} \, du \, dv.$$

We must show that using $\tilde{f}$ gives the same answer.

Proof. First, the coordinate-change formula gives

$$\tilde{g}(q) = D\phi(q)^\top g(\phi(q)) D\phi(q).$$

Taking determinants and then positive square roots,

$$\det \tilde{g}(q) = (\det D\phi(q))^2 \det g(\phi(q)),$$

$$\sqrt{\det \tilde{g}(q)} = |\det D\phi(q)| \sqrt{\det g(\phi(q))}.$$

Thus the area factor acquires exactly the absolute Jacobian determinant needed for a change of variables.

Write $x = (u, v)$, $\underline{q} = (r, s)$, and $x = \phi(q)$, with $dx = du dv$ and $dq = dr ds$. Using $\tilde{f} = f \circ \phi$ and the area-factor formula,

$$\begin{aligned} & \iint_{\tilde{U}} a(\tilde{f}(q)) \sqrt{\det \tilde{g}(q)} dq \\ &= \iint_{\tilde{U}} a(f(\phi(q))) \sqrt{\det g(\phi(q))} |\det D\phi(q)| dq \\ &= \iint_U a(f(x)) \sqrt{\det g(x)} dx. \end{aligned}$$

The last equality is the ordinary change-of-variables theorem for double integrals. This proves the lemma.

The function values and the area element must both be transformed:

$$\begin{aligned} a \circ \tilde{f} &= (a \circ f) \circ \phi, \\ \sqrt{\det \tilde{g}(q)} dq &= \sqrt{\det g(\phi(q))} |\det D\phi(q)| dq. \end{aligned}$$

The absolute value is essential: reversing the orientation of the parameters does not change a scalar surface integral.

The same argument applies to any integrable a , and on any overlap of surface coordinate patches. Taking $a = 1$ also proves that the area of a surface region is independent of its parametrization.